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Lead to owner: routing every inbound lead in minutes

Design notes for a workflow sample. I rebuilt the lead routing I designed at BaseN as an n8n workflow I can share. The rules, people, companies and data in the sample are made up; the results below are from the original.

The problem

At BaseN, new inbound leads sat in a shared queue for about a day before anyone owned them. About 40 new duplicates appeared every month, and only 55% of the key fields on a record were filled, so leads were slow to route and hard to report on.

What I built at BaseN, and what it changed

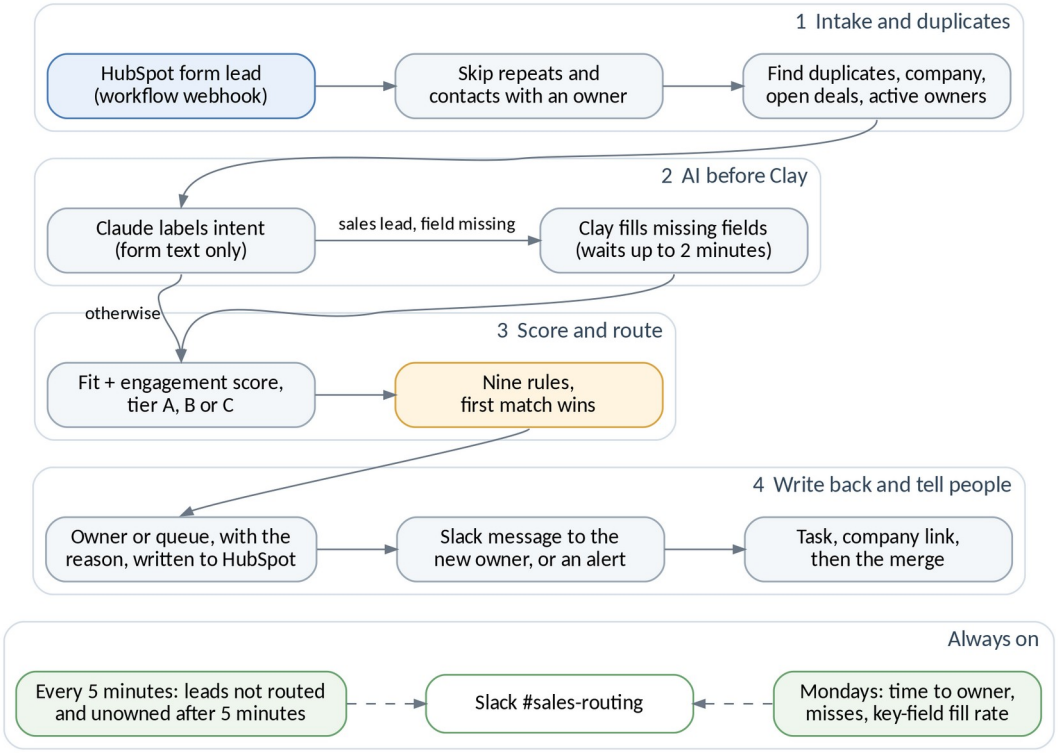
Nine routing rules across four regions, covering accounts that span two regions, owners who have left and existing customers, with a fit and engagement lead score feeding the rules. Dedupe on email for contacts and on domain for companies. Clay enrichment when a record is created. An LLM step that classified leads. I built it on HubSpot workflows, custom code actions (JavaScript) and Clay webhooks, with some steps in Zapier.

Measure (HubSpot reports)	Before	After
Time from create date to owner-assigned date	About a day, in a shared queue	**Under 5 minutes**
New duplicates	40 a month	**Under 5 a month**
Fill rate of a named set of key fields	55%	**90%**

It was one of 23 CRM automations I built that together save the sales team **about 35 hours a month**.

The rebuild: how a lead moves

The sample runs the same design in n8n, with HubSpot, Clay, Claude and Slack. Every Code node is built from tested JavaScript modules, so the workflow file runs the code the tests cover.



The nine rules, first match wins

#	Rule	Who gets the lead
1	Not a sales lead	Throwaway email, or support, job seeker, vendor pitch or spam: a queue, never a rep
2	Existing customer	The account owner
3	Open deal	The owner of the latest open deal at the company
4	Global account	An account that spans two regions keeps its named owner, matched on any of its domains
5	Company owner	A company already owned by an active rep stays with that rep
6	Owner has left	The regional team lead, with a task to reassign the company and an alert
7	Partner	Partner and reseller enquiries go to the partner manager
8	Region round robin	The rep whose last round-robin lead is oldest; tier A to senior reps first; reps on leave skipped
9	Routing queue	Country unknown or outside the regions, or nobody free: the routing queue, with an alert

A guard sits before rules 2 to 9: a lead that turns out to be a known contact with an active owner keeps that owner. Every lead gets a rule code and a plain-language reason on the record, for example **"The company's owner has left: routed to the regional team lead, with a task to reassign the company."**

Decisions that matter

- **The AI only labels the lead.** Claude returns intent, product interest, urgency, a confidence and a one-line summary for the rep, but not the owner. The answer is checked against fixed lists; below 0.7 confidence, keyword rules take over and the lead is flagged for review. If the API is down, the lead is still routed.
- **Only what the model needs.** It sees the form name, job title, company, website and message, not the name, email or phone. Angle brackets in that text are escaped and links are cut from the summary; a test shows an obeyed injection still can't pick the owner.
- **Exact merges only.** HubSpot already refuses a second contact with the same email, so the duplicates that get through are variants such as +tags and Gmail dots. Those merge on a normalized email key. A company found by name only can decide the owner, flagged for review, but is not linked automatically.
- **AI before Clay.** Support requests, job seekers and spam don't spend enrichment credits, and Clay is asked only for the fields that are missing. CRM values come first because people curate them.
- **A round robin with no counter.** The next lead goes to the rep whose last round-robin lead is oldest, read from HubSpot, so reps joining or back from leave need no special handling.
- **Owners are left alone.** An owner is set only when a rule picked one, and a contact that has one keeps it, even if a rep took it mid-run.

How it is tested and taken live

- **95 automated tests.** Each rule and the main edge cases; 22 made-up leads replayed against an in-memory HubSpot that answers the real search requests; and the n8n file itself run lead by lead against the tested code.
- **Shadow mode first.** The workflow records the owner it would choose without assigning, merging or messaging reps. Compare its choices with manual assignment, review the differences rule by rule, then switch to live.
- **Built-in measurement.** Every lead carries its intake and routing times and a key-field completeness score, the two measures from the BaseN reports. A watchdog reports leads never routed and unowned after 5 minutes, each once, and the week's numbers go to Slack on Mondays.

What I would watch at higher volume

- HubSpot search results trail writes by a few seconds, so two leads seconds apart can go to the same rep; the rotation then carries on as normal.
- HubSpot search allows five requests a second per account and each lead makes four searches, so every call retries; for bigger bursts, cap parallel runs with n8n's concurrency limit.
- Contacts created before go-live need a one-off backfill of the email key before their variants can be caught as duplicates.

Code, tests and the n8n workflow: github.com/Maliha-Tasnim/lead-to-owner. Nothing in it is copied from BaseN.